Class 22: Fixed Income II

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Today's Question

Silicon Valley Bank failed because of interest rate risk on “safe” Treasury bonds. What did they get wrong?

Positioning

Where We've Been	Where We Are	Where We're Going
Duration and interest rate risk	The yield curve; term structure theories	Managing fixed income portfolios

Matt Levine Reading

“Startup Bank Had a Startup Bank Run”

Key themes: held-to-maturity bonds, duration mismatch, and how a “safe” Treasury portfolio brought down a bank

Learning Objectives

By the end of today's class, you should be able to:

1. Interpret the yield curve and describe its typical shapes
2. Explain expectations hypothesis vs. liquidity preference vs. segmented markets
3. Describe how SVB's balance sheet made it vulnerable

Forward Rates

- We can extend this to the general case, solving for the forward rate from t to T:

$$(1+r_T)^T = (1+r_t)^t (1+f_{t,T})^{T-t}$$

$$f_{t,T} = \left[\frac{(1+r_T)^T}{(1+r_t)^t}\right]^{\frac{1}{T-t}} - 1$$

For example, $r_1 = 5\%$, $r_2 = 6\%$:

$$f_{1,2} = \frac{(1.06)^2}{1.05} - 1 = 7.01\%$$

Forward Rates and Term Structure

- Suppose investors are risk-neutral (or the future is certain)
- Then, forward rates should be equal to the expectations of interest rates (why? — what trade could you do if not?)
  - $f_{1,2} = E(r_{1,2})$
  - $f_{2,3} = E(r_{2,3})$
- Hence, long-term rates will represent a geometric average of current and future expected short rates

$$(1 + r_2)^2 = (1 + r_{0,1})(1 + f_{1,2}) = (1 + r_{0,1})E(1 + r_{1,2})$$

Forward Rates and Term Structure

- Now what if the investor is **risk-averse** and investing over two periods
- Faces a choice between a *sure* two-period rate, or a roll-over of a short-term bond
- Recall that a risk-averse investor prefers a sure thing if the expected returns are equal
- In equilibrium, two-period investors bid down the long-term rate:

$$(1+r_2)^2 = (1+r_{0,1})(1+f_{1,2}) < (1+r_{0,1})E(1+r_{1,2})$$

Forward Rates and Term Structure

- More commonly, we consider **risk-averse** *one-period* investor (needs liquidity at $t = 1$)
- Face a similar choice: a *sure* one-period rate, or a two-period bond that they sell in period 1

  1. Buy 1-year bond: $(1 + r_1)$
  2. Buy 2-year bond and sell at period 1 at price of $\frac{(1+r_2)^2}{1+E(r_{1,2})}$
     - Risky position is now flipped
- Recall that a risk-averse investor prefers a sure thing if the expected returns are equal
- In equilibrium, one-period investors bid down the short-term rate:

$$(1 + r_2)^2 = (1 + r_{0,1})(1 + f_{1,2}) > (1 + r_{0,1})E(1 + r_{1,2})$$

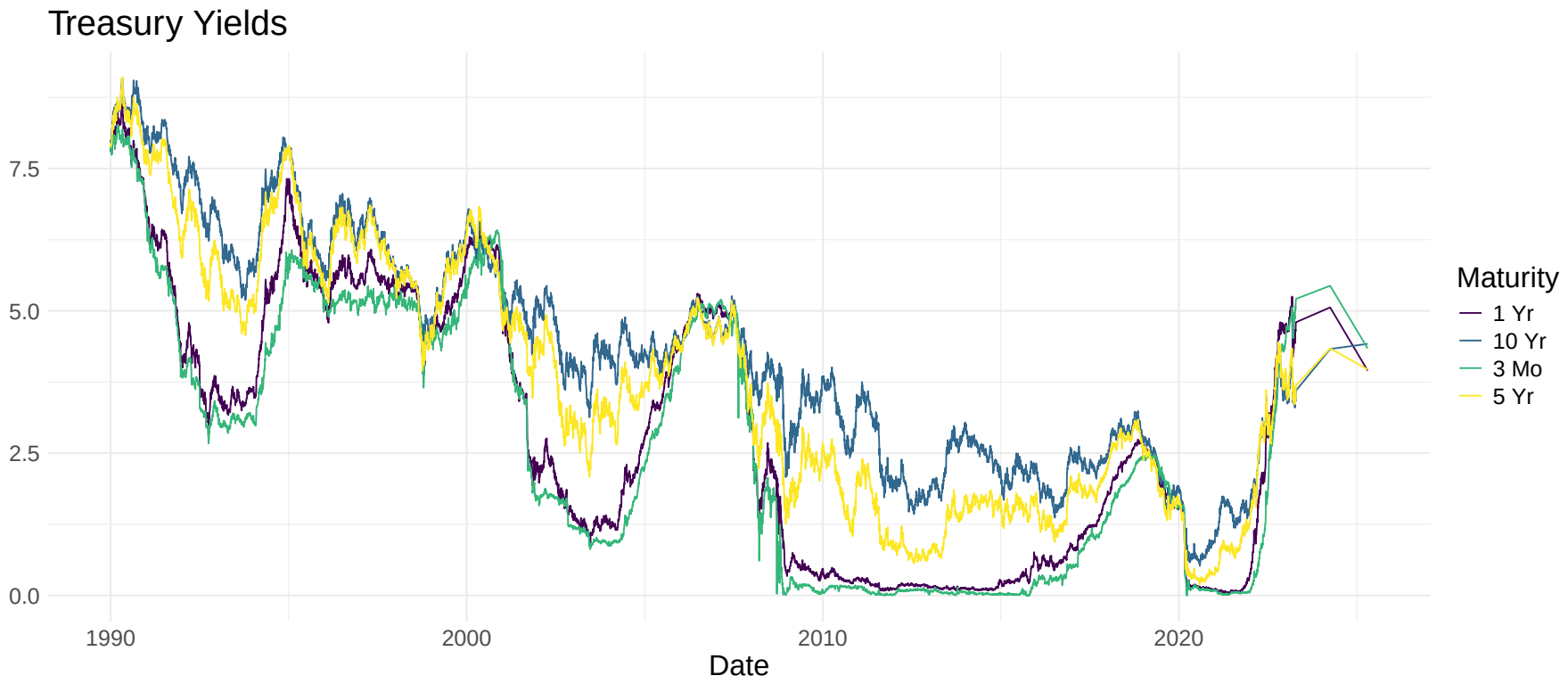
Interpreting Yield Curves

- Commonly, upward sloping yield curves are interpreted as evidence of:
  1. Higher future short rates
  2. Liquidity preference for short-term bonds
- Define the liquidity premium as the difference between the forward rate and the expected future short rate

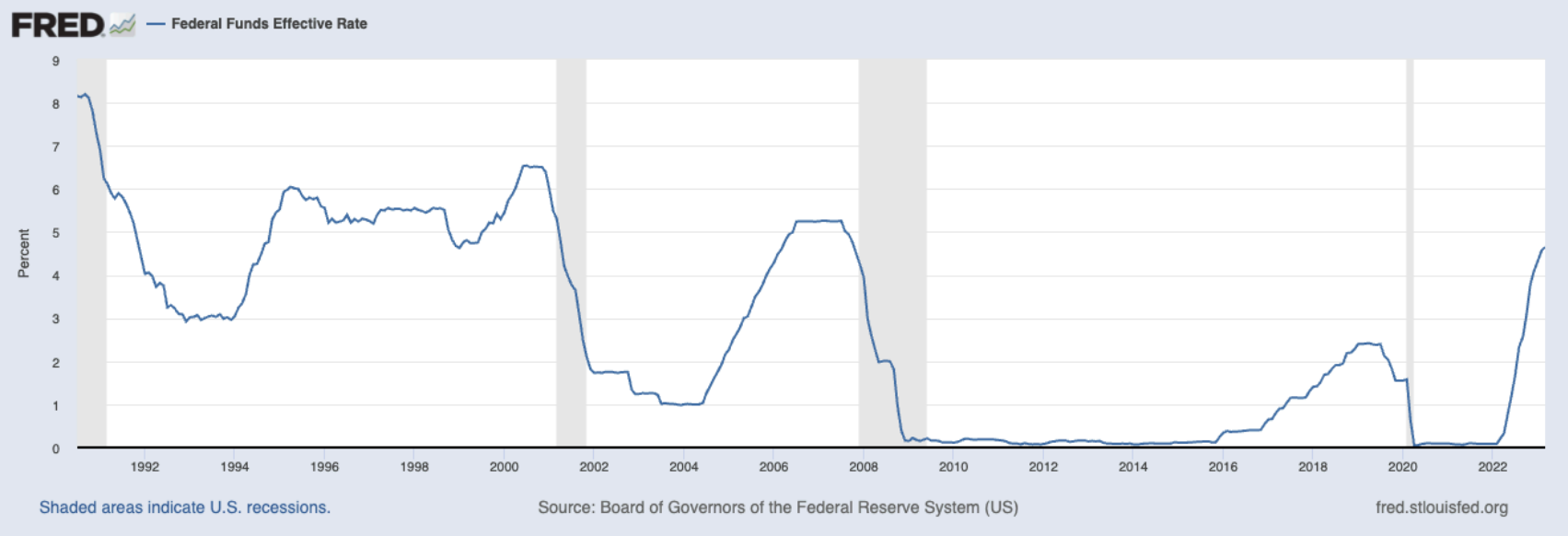
$$K = \left(1 + f_{t,t+1}\right) - E\left(1 + r_{t,t+1}\right)$$

- The larger this is, the more that you pay for "certainty" of next period's short-rate
- Can generate common yield curve patterns under different expectations and values of K

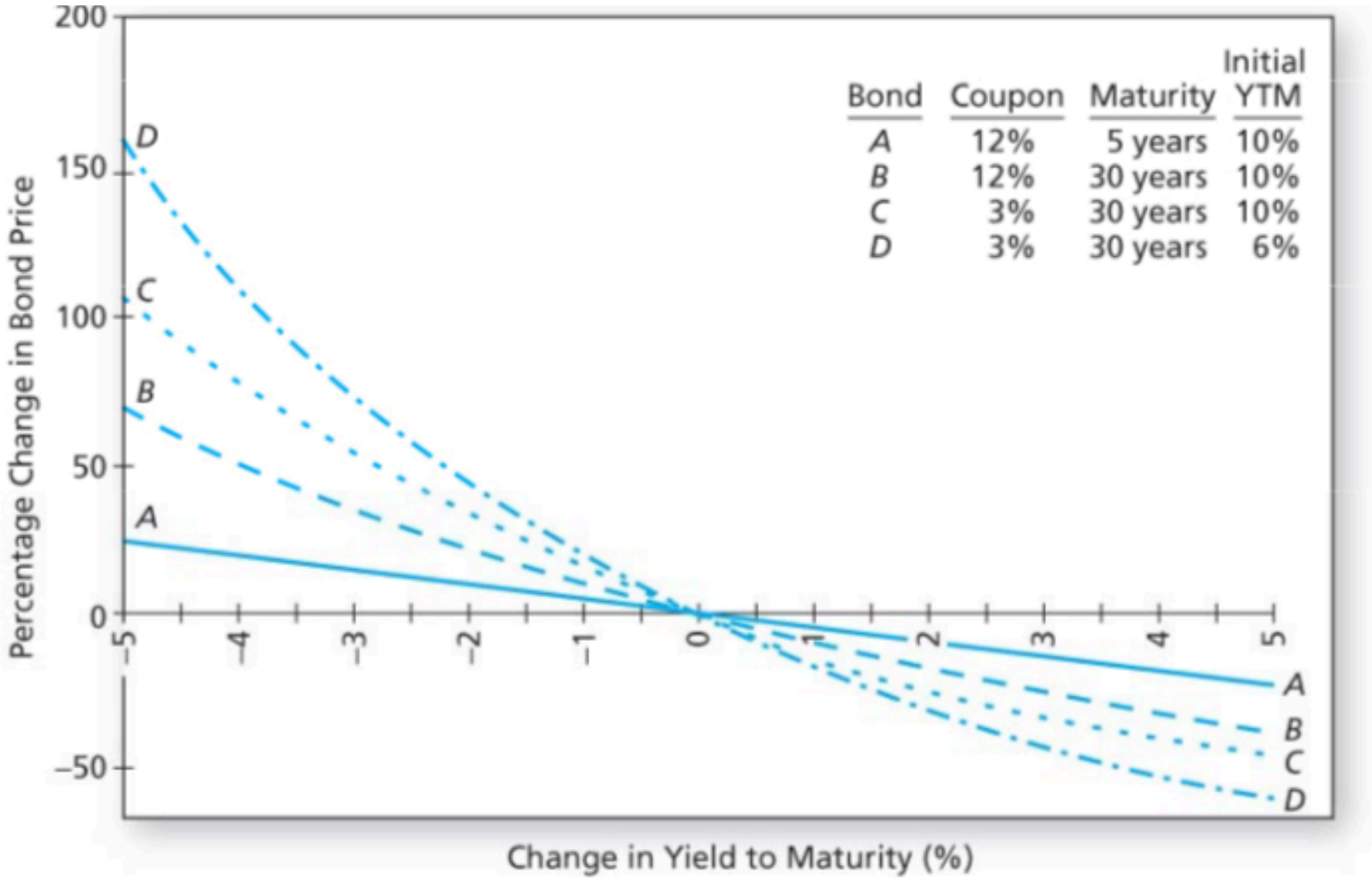
Treasury Yields Over Time



FRED: Treasury Yield Curve



Δ Bond Price and Δ Yield to Maturity



Bond Pricing Relationships

1. Bond prices and yields are inversely related.
2. Long-term bonds tend to be more price sensitive than short-term bonds.
3. Price sensitivity is inversely related to the bond's coupon rate.
4. Price sensitivity is inversely related to the yield to maturity at which the bond is selling.
5. An increase in a bond's yield to maturity results in a smaller price change than a decrease of equal magnitude.

Duration

- Need a single measure that tells us the interest rate sensitivity of a bond/portfolio
- **Duration** is a measure of the effective maturity
- The weighted average of the amount of time until each payment is received, with the weights proportional to the present value of the payment
- Duration is shorter than maturity for all bonds except zero coupon bonds
- Duration is equal to maturity for zero coupon bonds

Duration: Calculation

- Define duration as follows:

$$D = \sum_{t=1}^{T} t \times w_t$$

$$w_t = \frac{\left[C \frac{F_t}{(1+\text{YTM})^t}\right]}{P}$$

where YTM is the yield to maturity, P is the price, and CF_t is the cashflows at time t

- Can show that

Duration: Calculation

$$\frac{\Delta P}{P} = -D\frac{\Delta(1+\mathrm{YTM})}{1+\mathrm{YTM}}$$

Modified Duration

- Alternatively, *modified* duration (D^*) divides duration by $(1 + \mathrm{YTM})$ such that

$$\frac{\Delta P}{P} = -D^* \Delta \ \mathrm{YTM}$$

Next Class: April 23

- Topics: Fixed Income III
  - Duration matching
  - Orange County case
  - Interest rate swaps

- Matt Levine reading: TBD